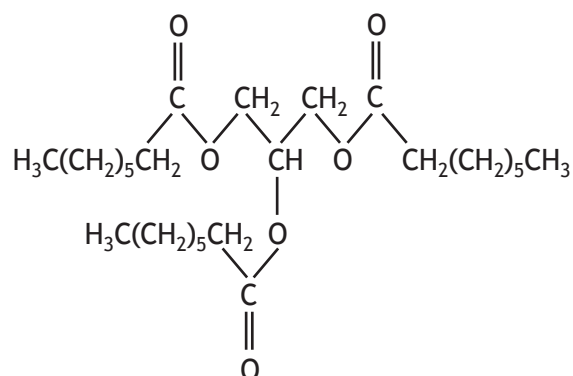


3. Cheese is a complex substance containing a wide variety of chemicals.

(a) The structure of a fat found in cheese is shown below.



(i) (A) The alcohol needed to form fat molecules is glycerol.

State the systematic name for glycerol.

1

(B) Name the type of reaction used to form fat molecules from fatty acids and glycerol.

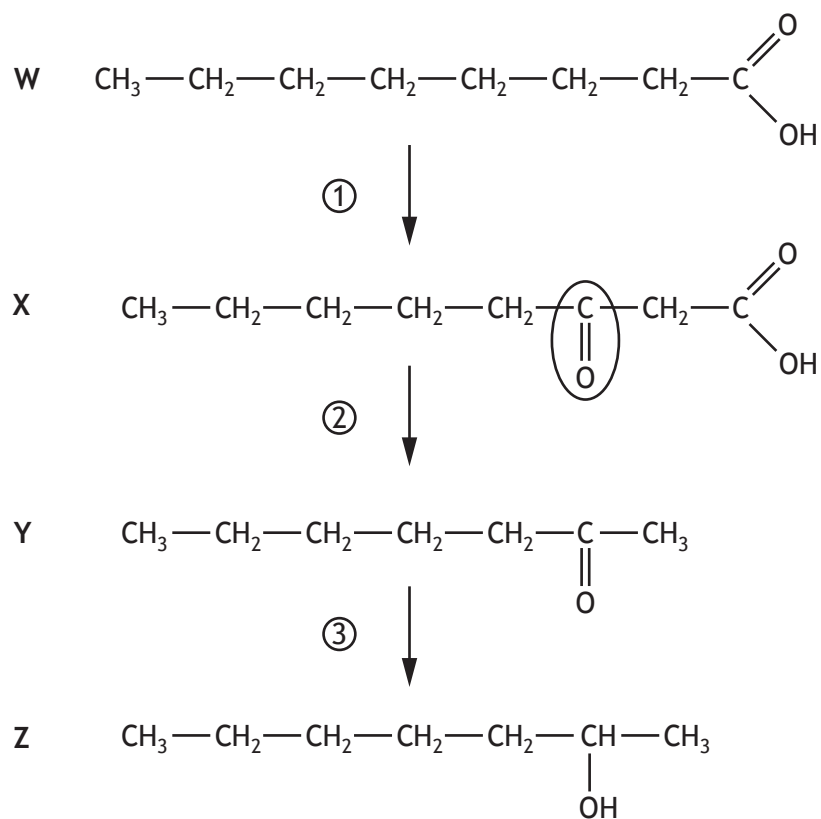
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* X 8 1 3 7 6 0 1 0 8 *

3. (a) (continued)

(ii) Fatty acid **W** reacts as shown.



- (A) Identify the functional group circled in molecule **X**. 1
- (B) Name molecule **Y**. 1
- (C) Identify the type of reaction used to convert molecule **Y** into molecule **Z**. 1
- (D) State which of the reactions, ①, ② or ③, results in an increase in the oxygen to hydrogen ratio. 1

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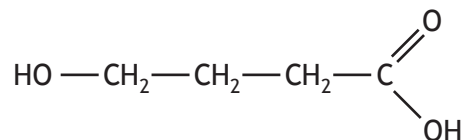


* X 8 1 3 7 6 0 1 0 9 *

3. (a) (continued)

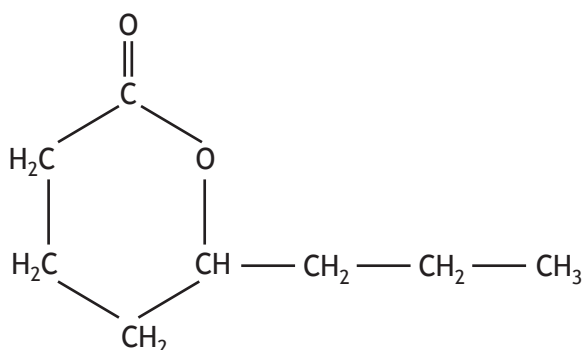
(iii) Fatty acids can be converted into hydroxycarboxylic acids.

An example of a hydroxycarboxylic acid is shown.



The two functional groups in a hydroxycarboxylic acid react together to form a cyclic ester.

An example of a cyclic ester is shown.



Draw a structural formula for the hydroxycarboxylic acid that can be used to produce this cyclic ester.

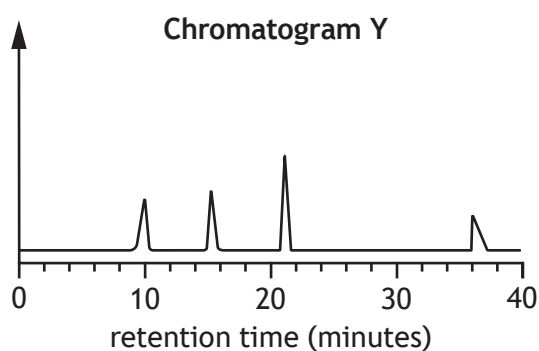
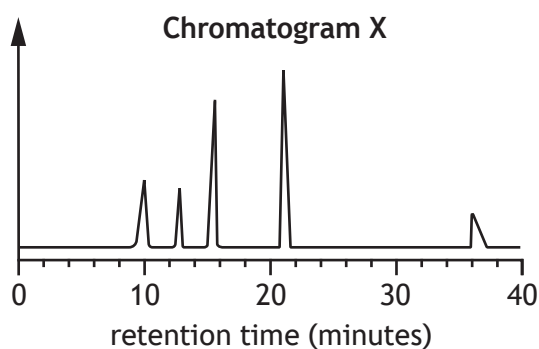
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3. (a) (continued)

- (iv) The flavour of cheese changes over time as the concentrations of flavour molecules change.

Gas chromatography can be used to analyse the concentrations of flavour molecules.

(A) Chromatograms for two samples of cheese are shown below.



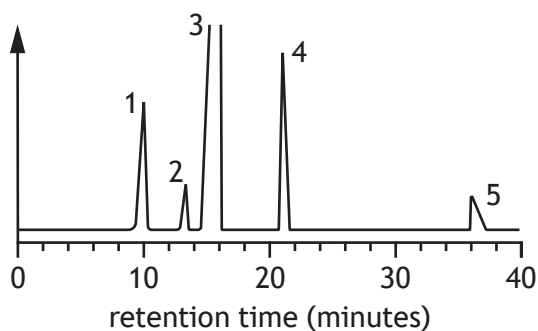
Determine the retention time, in minutes, of the peak in **Chromatogram X** that is missing in **Chromatogram Y**.

1

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3. (a) (iv) (continued)

- (B) The following chromatogram was obtained from another sample of cheese. The concentration of a flavour molecule in cheese can be determined by calculating the area under the peak that corresponds to that molecule.



The concentration of flavour molecule 3 cannot be determined from this chromatogram.

Suggest what would need to be done to the sample to allow the concentration of flavour molecule 3 to be determined.

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3. (b) (continued)

(iii) When cheese is heated, the proteins change shape.

State the term used to describe this process.

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(c) Processed cheese is made from cheese, soluble milk proteins, water and an emulsifier.

State the function of an emulsifier.

1

(d) A compound added to cheese as a mould inhibitor has the formula $\text{Ca}^{2+}(\text{CH}_3\text{CH}_2\text{COO}^-)_2$.

Name this compound.

1

